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United States Patent	12418738
Kind Code	B2
Date of Patent	September 16, 2025
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Photoelectric conversion apparatus, photoelectric conversion system, and photoelectric conversion method for shading correction

Abstract

Photoelectric conversion apparatus includes read circuit for reading signals of pixel array, and signal processing unit for generating pixel signal by performing CDS on signals read by the read circuit and performing shading correction on the pixel signal. The read circuit has function of reading signal of the pixel array with first and second gain. The read circuit reads noise level with the first gain and reads optical signal level with the first or second gain from selected effective pixel for the correlated double sampling. In the shading correction, pixel signal of effective pixel from which optical signal level has been read with the first gain is corrected based on first correction value, and pixel signal of effective pixel from which optical signal level has been read with the second gain is corrected based on second correction value.

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Appl. No.: 18/453342

Filed: August 22, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20240080590 A1	Mar. 07, 2024

Foreign Application Priority Data

JP	2022-140208	Sep. 02, 2022
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Publication Classification

Int. Cl.: H04N25/78 (20230101); H04N25/51 (20230101); H04N25/61 (20230101); H04N25/616 (20230101); H04N25/677 (20230101)

U.S. Cl.:

CPC H04N25/78 (20230101); H04N25/51 (20230101); H04N25/61 (20230101); H04N25/616 (20230101); H04N25/677 (20230101);

Field of Classification Search

CPC: H04N (25/78); H04N (25/61); H04N (25/616); H04N (25/51); H04N (25/677)

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Background/Summary

BACKGROUND OF THE INVENTION

Field of the Invention

(1) The present invention relates to a photoelectric conversion apparatus, a photoelectric conversion system, and a photoelectric conversion method.

Description of the Related Art

(2) A photoelectric conversion apparatus is demanded to have a large number of pixels and a high frame rate. In order to satisfy the demand, one of challenges is to increase the readout speed of signals by the photoelectric conversion apparatus. In addition to increasing the readout speed, there is also a strong demand for improvement of the S/N ratio and expansion of the dynamic range.

There is known a photoelectric conversion apparatus that performs analog-to-digital (A/D) conversion, using different A/D converter circuits, of a first pixel signal obtained by amplifying a pixel signal with a first gain and a second pixel signal obtained by amplifying the pixel signal with a second gain different from the first gain. The photoelectric conversion apparatus selectively outputs one of the A/D-converted first pixel signal and the A/D-converted second pixel signal in accordance with the level of the pixel signal. With this, it is possible to implement expansion of the dynamic range and improvement of the S/N ratio. Japanese Patent Laid-Open No. 2014-140152 discloses a technique of correcting the offset error and the gain error generated when selectively outputting the first pixel signal and the second pixel signal.

(3) It is known that the S/N ratio is improved by a Correlated Double Sampling (CDS) process in which the optical signal level and noise level generated by photoelectric conversion are read out from a pixel, and a pixel signal corresponding to the difference between the optical signal level and the noise level is generated. However, if the gain used when reading out the optical signal level is different from the gain used when reading out the noise level, a noise component can remain in the pixel signal generated by the CDS process.

SUMMARY OF THE INVENTION

(4) The present invention provides a technique advantageous in effectively reducing a noise component from a pixel signal.

(5) One of aspects of the present disclosure provides a photoelectric conversion apparatus comprising a pixel array including a plurality of effective pixels, a readout circuit configured to read out a signal of the pixel array, and a signal processing unit configured to generate a pixel signal by performing correlated double sampling on signals read out by the readout circuit from an effective pixel selected from the plurality of effective pixels, and perform shading correction on the pixel signal, wherein the readout circuit has a function of reading out a signal of the pixel array with a first gain, and a function of reading out a signal of the pixel array with a second gain different from the first gain, the readout circuit reads out a noise level with the first gain and reads out an optical signal level with the first gain or the second gain from an effective pixel selected from the plurality of effective pixels for the correlated double sampling, and in the shading correction, a pixel signal of an effective pixel from which an optical signal level has been read out with the first gain is corrected based on a first correction value, and a pixel signal of an effective pixel from which an optical signal level has been read out with the second gain is corrected based on a second correction value.

(6) Further features of the present invention will become apparent from the following description of exemplary embodiments with reference to the attached drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a view exemplarily showing the arrangement of a photoelectric conversion apparatus according to the first embodiment;

- (2) FIG. 2 is a view exemplarily showing the arrangement of a pixel;
- (3) FIG. 3 is a view for explaining a basic readout operation of the photoelectric conversion apparatus according to the first embodiment;
- (4) FIGS. 4A and 4B are views for explaining the arrangement and principle of A/D conversion;
- (5) FIG. 5 is a view exemplarily showing the operation of a column circuit in the first embodiment;
- (6) FIG. 6 is a view exemplarily showing the relationship between an optical signal level and the result of A/D conversion in a case in which the change rate (slope) of a reference signal is changed in accordance with the optical signal level;
- (7) FIG. 7 is a view exemplarily showing the arrangement of a pixel array in the first embodiment;
- (8) FIGS. 8A and 8B are views exemplarily showing an operation of reading out a signal from a reference pixel region in a calibration operation according to the first embodiment;
- (9) FIG. 9 is a view exemplarily showing the arrangement of a signal processing unit in the first embodiment;
- (10) FIG. 10 is a view exemplarily showing the arrangement of a pixel array in the second embodiment;
- (11) FIG. 11 is a view exemplarily showing the arrangement of a signal processing unit in the second embodiment;
- (12) FIG. 12 is a view exemplarily showing the arrangement of a signal processing unit in the fourth embodiment;
- (13) FIG. 13 is a view exemplarily showing the arrangement of a column circuit in the fifth embodiment;
- (14) FIG. 14 is a view exemplarily showing the operation of the column circuit in the fifth embodiment;
- (15) FIGS. 15A and 15B are views exemplarily showing an operation of reading out a signal from a reference pixel region in a calibration operation according to the fifth embodiment;
- (16) FIG. 16 is a view exemplarily showing the operation of a photoelectric conversion apparatus according to the sixth embodiment; and
- (17) FIG. 17 is a view exemplarily showing the structure of the photoelectric conversion apparatus.
- (18) FIG. 18 is a view exemplarily showing the arrangement of an image capturing apparatus.

DESCRIPTION OF THE EMBODIMENTS

(19) Hereinafter, embodiments will be described in detail with reference to the attached drawings. Note, the following embodiments are not intended to limit the scope of the claimed invention. Multiple features are described in the embodiments, but limitation is not made to an invention that requires all such features, and multiple such features may be combined as appropriate. Furthermore, in the attached drawings, the same reference numerals are given to the same or similar configurations, and redundant description thereof is omitted.

(20) Hereinafter, with reference to FIGS. 1 to 9, a photoelectric conversion apparatus **100** according to the first embodiment of the present invention will be described below. The photoelectric conversion apparatus **100** may be composed of a chip formed by one semiconductor layer or semiconductor substrate, may be composed of a chip formed by a plurality of semiconductor layers or semiconductor substrates, or may be composed of a plurality of chips. The photoelectric conversion apparatus **100** can be formed as an image sensor or an apparatus including an image sensor.

(21) FIG. 1 exemplarily shows the arrangement of the photoelectric conversion apparatus **100** according to the first embodiment. The photoelectric conversion apparatus **1100** can include a pixel array **101** which includes a plurality of pixels **102** arranged so as to form a plurality of rows and a plurality of columns. In order to distinguish the pixel **102** from a reference pixel and OB pixel to be described later, the pixel **102** can also be referred to as an effective pixel. The pixel array **101** can include a plurality of column signal lines **110**. Each column of the pixel array **101** can be assigned with at least one column signal line **110**. The photoelectric conversion apparatus **100** can also

include a readout circuit **105** that reads out signals of the pixel array **101**. The readout circuit **105** can have a function of reading out a signal of the pixel array **101** with a first gain, and a function of reading out a signal of the pixel array **101** with a second gain different from the first gain. The readout circuit **105** may be understood to have a function of reading out signals of the pixel array **101** with a plurality of gains different from each other. The readout circuit **105** can include a plurality of column circuits, and each column circuit can be configured to read out signals of the pixel array **101** through at least one column signal line **110**.

(22) The photoelectric conversion apparatus **100** can include a signal processing unit **108** that performs a correlated double sampling process (CDS process) and a correction process on signals read out from the pixels **102** (effective pixels) in the pixel array **101** by the readout circuit **105**. The photoelectric conversion apparatus **100** can further include a vertical scanning unit **103**, a control unit **104**, and a horizontal scanning unit **107**. In an arrangement example, the signal processing unit **108** can form a part of the photoelectric conversion apparatus **100**. In another arrangement example, the signal processing unit **108** does not form a part of the photoelectric conversion apparatus **100**. In this case, a system including the photoelectric conversion apparatus **100** and the signal processing unit **108** can be understood as a photoelectric conversion system. In such a photoelectric conversion system, the function of the signal processing unit **108** may be provided by a computer such as a personal computer, may be provided by a processor such as an ASIC, or may be implemented by another arrangement.

(23) In the example shown in FIG. **1**, the plurality of pixels **102** forming the pixel array **101** are arranged so as to form a matrix of m columns and n rows. The pixel unit **102** can also be written as a pixel $P(i, j)$ to indicate its position. i indicates the column where the pixel $P(i, j)$ is arranged, and j indicates the row where the pixel $P(i, j)$ is arranged. The vertical scanning unit **103** selects one row from n rows by n sets of row selection lines **109** ($V(j)$ ($j=1$ to n)). Selecting a row means selecting m pixels **102** arranged in the row. The signals of the pixels **102** arranged in the selected row can be read out by the readout circuit **105** via the column signal lines **110** ($H(i)$ ($i=1$ to m)). Readout by the readout circuit **105** includes a process of converting (that is, A/D-converting) a plurality of analog signals output from the pixels **102** to the column signal lines **110** into a plurality of digital signals in parallel, and the plurality of digital signals are temporarily stored in a memory unit **106**. The readout circuit **105** may include an amplification circuit and the like. The amplification circuit can amplify the signal output from the pixel **102**. The plurality of digital signals stored in the memory unit **106** can be then sequentially selected by the horizontal scanning unit **107** and supplied to the signal processing unit **108**. The signal processing unit **108** processes the digital signals supplied as described above, and outputs signals obtained by the process. The control unit **104** can acquire setting information for setting the image capturing condition and the like in image capturing by the photoelectric conversion apparatus **100**, and supply a control signal corresponding to the image capturing condition to each component included in the photoelectric conversion apparatus **100**. The control unit **104** can be configured to control the vertical scanning unit **103**, the readout circuit **105**, the memory unit **106**, the horizontal scanning unit **107**, and the signal processing unit **108** based on the setting information.

(24) FIG. **2** shows an arrangement example of the pixel **102**($P(i, j)$). The pixel **102** can include, for example, a photoelectric conversion element **2201**, a transfer transistor **202**, a floating diffusion FD, a reset transistor **203**, an amplification transistor **204**, and a selection transistor **205**. The photoelectric conversion element **201** performs photoelectric conversion, thereby generating electric charges corresponding to the incident light amount. The photoelectric conversion element **201** is, for example, a photodiode. The transfer transistor **202** transfers the electric charges generated by photoelectric conversion in the photoelectric conversion element **201** to the floating diffusion FD. The electric charges transferred to the floating diffusion FD are converted into a potential by an electrostatic capacitance included in the floating diffusion FD. The amplification transistor **204** outputs a signal corresponding to the potential of the floating diffusion FD to the

column signal line **110**. The reset transistor **203** resets the potential of the floating diffusion FD to a predetermined potential. One row selection line **109** (V(j)) described above includes a transfer control line PTX(j), a reset control line PRES(j), and a row selection line PSEL(j) used to control the transfer transistor **202**, the reset transistor **203**, and the selection transistor **205**, respectively. Note that a transfer control signal, a reset control signal, and a row selection signal respectively supplied to the transfer control line PTX(j), the reset control line PRES(j), and the row selection line PSEL(j) are also identified as PTX(j), PRES(j), and PSEL(j), respectively.

(25) Next, with reference to FIG. 3, a basic readout operation in the photoelectric conversion apparatus **100** will be described. FIG. 3 exemplarily shows a timing chart of the operation of reading out signals from the pixel array **101**. Here, the operation of reading out signals from the pixel **102** in the jth row by the vertical scanning unit **103** is representatively shown. Note that signals shown in FIG. 3 are high-active signals.

(26) At time t₃₀₁, a horizontal synchronization signal SYNC is activated to high level, and the row selection signal PSEL(j) of the jth row is activated to high level. If the row selection signal PSEL(j) is set at high level, the selection transistor **205** of the pixel **102** in the selected row is set in the ON state, and the pixel **102** in the selected row is electrically connected to the column signal line **110**.

(27) At time t₃₀₂, the reset control signal PRES(j) is activated to high level. This sets the reset transistor **203** of the pixel **102** in the ON state, and the floating diffusion FD is reset to the reset potential corresponding to a power supply potential VDD.

(28) At time t₃₀₃, the reset control signal PRES(j) is inactivated to low level, and the reset transistor **203** is set in the OFF state. Thus, the reset of the potential of the floating diffusion FD is released. Since the selection transistor **205** is kept in the ON state, a signal corresponding to the gate potential of the amplification transistor **204** at the time of release of the reset of the potential of the floating diffusion FD is output to the column signal line **110**. In the period from time t₃₀₃ to time t₃₀₆, a noise level (N level) is output from the pixel **102** to the column signal line **110**.

(29) In the period from time t₃₀₄ to time t₃₀₅, the noise level output to the column signal line **110** can be read out as a digital signal by the readout circuit **105**. The digital signal of the noise level read out by the readout circuit **105** is stored in the memory unit **106**. The operation performed in the period from time t₃₀₄ to time t₃₀₅, that is, the operation of converting the noise level into the digital signal is referred to as N conversion.

(30) At time t₃₀₆, the transfer control signal PTX(j) is activated to high level. This sets the transfer transistor **202** of the pixel **102** in the ON state, and electric charges generated by photoelectric conversion in the photoelectric conversion element **201** are transferred to the floating diffusion FD. A signal corresponding to the electric charges generated in the photoelectric conversion element **201** is output to the column signal line **110**. In the period from time t₃₀₆ to time t₃₁₀, an optical signal level (S level) is output from the pixel **102** to the column signal line **110**.

(31) At time t₃₀₇, the transfer control signal PTX(j) is inactivated to low level. This sets the transfer transistor **202** in the OFF state. Even after the transfer transistor **202** is set in the OFF state, the optical signal level continues to be output to the column signal line **110**.

(32) In the period from time t₃₀₈ to time t₃₀₉, the optical signal level output to the column signal line **110** is read out as a digital signal by the readout circuit **105**. The digital signal of the optical signal level read out by the readout circuit **105** is stored in the memory unit **106**. The operation performed in the period from time t₃₀₈ to time t₃₀₉, that is, the operation of converting the optical signal level into the digital signal is referred to as S conversion.

(33) In the period from time t₃₁₀ to time t₃₁₁, in accordance with a horizontal scanning pulse signal output from the horizontal scanning unit **107**, a pair of the noise level and the optical signal level held in the memory unit **106** is output from the memory unit **106** to the signal processing unit **108**. By repeating the horizontal scanning until the last column, readout of pairs of the noise levels and the optical signal levels of the pixels **102** for one row arranged in the readout target row is completed. The signal processing unit **108** includes a CDS circuit that performs a CDS process of

generating a pixel signal by subtracting the noise level from the optical signal level. The CDS circuit outputs a pixel signal having undergone the CDS process.

(34) Next, with reference to FIGS. 4A and 4B, the arrangement and principle of A/D conversion performed by the readout circuit **105** will be exemplarily described. The readout circuit **105** includes, for each column, a column circuit **400** including an A/D converter, and the A/D converter can include a comparator **401**, a counter/latch circuit **402**, and a ramp circuit **403** as exemplarily shown in FIG. 4A. The ramp circuit **403** is a circuit that generates or outputs a reference signal VRAMP (ramp signal) which changes at a constant change rate (time change rate), that is, changes with a constant slope. The comparator **401** compares a signal VIN output from the pixel array **101** to the column signal line **110** with the reference signal VRAMP output from the ramp circuit **403**, and outputs the comparison result.

(35) As exemplarily shown in FIG. 4B, prior to the start of readout of the signal VIN output from the pixel array **101**, the operation of the comparator **401** is started (time **t4010**). Once the signal VIN output from the pixel array **101** stabilizes, the count value of the counter/latch circuit **402** is reset at time **t4011**. In synchronization with the timing of reset of the count value of the counter/latch circuit **402**, the level of the reference signal VRAMP output from the ramp circuit **403** increases along with the time elapse from time **t4011**. If the level of the reference signal VRAMP output from the ramp circuit **403** exceeds the optical signal level of the signal VIN output from the pixel array **101**, the output of the comparator **401** is inverted (time **t4012**). The counter/latch circuit **402** performs a count operation in the period (time **t4011** to time **t4012**) from the reset of the count value to the inversion of the output of the comparator **401**. With this operation, the count value proportional to the level of the signal output from the pixel array **101** can be obtained, and the obtained count value serves as the result of A/D conversion. Note that the method of comparing the signal from the pixel array **101** with the reference signal, the comparison method by the counter/latch circuit, and the like described here are merely examples, and other methods may be used as long as the period from the reset of the count value to the inversion of the output of the comparator **401** can be detected.

(36) FIG. 5 exemplarily shows the operation of one column circuit **400** in the readout circuit **105** according to the first embodiment. In FIG. 5, the abscissa exemplarily represents time, the ordinate in the upper stage exemplarily represents the level of the reference signal and the level of the signal from the pixel array **101**, and the ordinates in the lower stage exemplarily represents the output of the comparator **401**. Here, an example will be described in which the change rate (slope) of the reference signal VRAMP output from the ramp circuit **403** is changed in accordance with the level of the signal VIN output from the pixel array **101**. The ramp circuit **403** can selectively generate or output a first reference signal VRAMP-1 which changes at a first change rate, a second reference signal VRAMP-2 which changes at a second change rate whose change rate (slope) is larger than the first range rate, and a determination reference signal.

(37) First, in order to perform A/D conversion of the noise level (N level), the operation of the comparator **401** is started at time **t501**. The count of the counter/latch circuit **402** is reset at time **t502**, and the level of the reference signal VRAMP output from the ramp circuit **403** is changed at the first change rate. Since the noise level is small, the first reference signal VRAMP-1 (first ramp signal) having a small slope is used for A/D conversion of the noise level. The counter/latch circuit **402** performs the count operation in the period (time **t502** to time **t503**) from the reset of the count to the inversion of the output of the comparator **401**. With this, the noise level is A/D-converted.

(38) Then, in the level determination period, the column circuit **400** determines the optical signal level which is the signal corresponding to electric charges accumulated in the photoelectric conversion element **201** in the pixel **102** in the pixel array **101**. In the level determination period, the ramp circuit **403** outputs, to the comparator **401**, the determination reference signal with a determination level Vs as the maximum level. The determination level Vs is a threshold value for determination. The comparator **401** compares the signal VIN output from the pixel array **101** with

the determination reference signal. Here, the count value of the counter/latch circuit **402** is reset at time **t504**, and the ramp circuit **403** starts to output the determination reference signal. If the level of the signal VIN (optical signal level) output from the pixel array **101** is larger than the determination level V_s ($S > V_s$), the output of the comparator **401** is not inverted, so that the count value continues to increase until the level determination period ends at time **t506**. On the other hand, if the level of the signal VIN output from the pixel array **101** is smaller than the determination level V_s ($S < V_s$), for example, the output of the comparator **401** is inverted at time **t505**, so that the count value stops increasing. In this manner, based on the count value of the counter/latch circuit **402**, the column circuit **400** can determine whether the optical signal level is larger or smaller than the determination level V_s .

(39) The column circuit **400** supplies the result of determination as to whether the optical signal level is larger or smaller than the determination level V_s to the memory unit **106** as luminance determination information (luminance determination information LL to be described later). The luminance determination information is stored in the memory unit **106** while being associated with the digital signal generated by the column circuit **400**. The luminance determination information can be, for example, information which has a value 1 (=high) if the optical signal level (S) is larger than the determination level V_s ($S > V_s$), and has a value 0 (=low) if the optical signal level is smaller than the determination level V_s ($S < V_s$). Note that it may be controlled such that the timing of resetting the count value of the counter/latch circuit **402** is the time at which the output of the ramp circuit **403** stabilizes at the determination level V_s , and the count value becomes 0 if the optical signal level (S) is smaller than the determination level V_s ($S < V_s$).

(40) If the optical signal level (S) is smaller than the determination level V_s ($S < V_s$), from time **t507**, the optical signal level can be A/D-converted using the first reference signal VRAMP-1 as in A/D conversion of the noise level. With this, in the example shown in FIG. 5, the count value can be obtained in the period from time **t507** to time **t508**. On the other hand, if the optical signal level (S) is larger than the determination level V_s , the optical signal level can be A/D-converted using the second reference signal VRAMP-2 which changes at the second change rate that is a times the first change rate of the first reference signal VRAMP-1. With this, in the example shown in FIG. 5, the count value can be obtained in the period from time **t507** to time **t509**. Here, the first change rate of the first reference signal VRAMP-1 and the second change rate of the second reference signal VRAMP-2 can be understood as the gains of readout by the readout circuit **105**, and correspond to the first gain and the second gain, respectively. That is, readout of the signal from the pixel array **101** using the first reference signal VRAMP-1 corresponds to readout of the signal from the pixel array **101** with the first gain. Further, readout of the signal from the pixel array **101** using the second reference signal VRAMP-2 corresponds to readout of the signal from the pixel array **101** with the second gain. That is, the readout circuit **105** has a function of reading out a signal of the pixel array **101** with the first gain, and a function of reading out a signal of the pixel array **101** with the second gain different from the first gain.

(41) FIG. 6 exemplarily shows the relationship between the optical signal level and the result of A/D conversion in a case in which the change rate (slope) of the reference signal is changed in accordance with the optical signal level. The abscissa of FIG. 6 represents the optical signal level output from the pixel array **101**, and the ordinate represents the result (A/D-converted value) of A/D conversion of the optical signal level. The solid line indicates the digital value (A/D-converted value) having undergone A/D conversion by the comparator **401** and the counter/latch circuit **402** and supplied to the signal processing unit **108** via the horizontal scanning unit **107**. As has been described above, the optical signal level having a value smaller than the determination level V_s is A/D-converted using the first reference signal VRAMP-1, and the optical signal level having a value larger than the determination level V_s is A/D-converted using the second reference signal VRAMP-2. Therefore, as shown in FIG. 6, the optical signal level after A/D conversion becomes discontinuous before and after the determination level V_s .

(42) To solve this problem, the signal processing unit **108** multiplies the A/D-converted value of the optical signal level larger than the determination level V_s by a ratio α , which is the ratio (second change rate/first change rate) of the first change rate of the first reference signal VRAMP-1 and the second change rate of the second reference signal VRAMP-2.

(43) Subsequently, the principle of the basic reset noise removal process performed in the signal processing unit **108** will be described. The A/D-converted optical signal level described with reference to FIG. 5 includes not only the signal component corresponding to the electric charges accumulated in the photoelectric conversion element **201** of the pixel **102** in the pixel array **101**, but also a noise component such as reset noise due to the reset transistor **203**. On the other hand, the A/D-converted noise level described with reference to FIG. 5 includes a noise component such as reset noise due to the reset transistor **203**. Therefore, by performing the CDS process of subtracting the A/D-converted noise level from the A/D-converted optical signal level, it is possible to reduce the reset noise from the optical signal level.

(44) If the optical signal level is smaller than the determination level V_s , reset noise removal by the CDS process can be expressed as:

$$(SL+NL+DL(T2))-(NL+DL(T1))=SL+DL(T2)-DL(T1) \quad (1)$$

(45) In equation (1), SL is the optical signal level having undergone A/D conversion using the first reference signal VRAMP-1, and NL is the noise level having undergone A/D conversion using the first reference signal VRAMP-1. Further, in equation (1), each of DL(T2) and DL(T1) is a value obtained by converting, into an error component, the amount of response delay of the comparator **401** in a case of using the first reference signal VRAMP-1. The amount of response delay corresponds to the time from the timing at which the value of the signal output from the pixel array **101** exceeds the value of the first reference signal VRAMP-1 to the inversion of the output of the comparator **401** in response thereto. This amount of response delay appears as an error in the result of A/D conversion. DL(T) is expressed as a function of T because it has dependency on the time (T) of A/D conversion. T1 is the time when the noise level is A/D-converted using the first reference signal VRAMP-1, and T2 is the time when the optical signal level is A/D-converted using the first reference signal VRAMP-1. DL(T2)-DL(T1), which is the influence of the amount of response delay of the comparator **401**, cannot be reduced by the CDS process. Since DL(T2)-DL(T1) has dependency on the time T of A/D conversion, it appears as a contrast changing in the vertical direction (column direction) in an image output from the photoelectric conversion apparatus **100**. This is referred to as vertical shading. The vertical shading is corrected by the signal processing unit **108**.

(46) On the other hand, if the optical signal level is larger than the determination level V_s , reset noise removal by the CDS process can be expressed as:

$$\alpha \times (SH+NH+DH(T3))-(NL+DL(T1))=\alpha SH+\alpha DH(T3)-DL(T1) \quad (2)$$

(47) In equation (2), SH is the optical signal level having undergone A/D conversion using the second reference signal VRAMP-2, and NH is the noise level having undergone A/D conversion using the second reference signal VRAMP-2. Further, in equation (2), DH(T3) is a value obtained by converting, into an error component, the amount of response delay of the comparator **401** in a case of using the second reference signal VRAMP-2. The amount of response delay corresponds to the time from the timing at which the value of the signal output from the pixel array **101** exceeds the value of the second reference signal VRAMP-2 to the inversion of the output of the comparator **401** in response thereto. This amount of response delay appears as an error in the result of A/D conversion. Since αNH and NL of the left-hand side of equation (2) indicate the same noise level, it can be regarded that $NL=\alpha NH$. However, the reference signal for A/D conversion of the optical signal level and the reference signal for A/D conversion of the noise level are different, and the time of A/D conversion is also different. Therefore, $\alpha DH(T3)-DL(T1)$, which is the influence of the amount of response delay of the comparator **401**, cannot be reduced by the CDS process. Since $\alpha DH(T3)-DL(T1)$ has dependency on the time T of A/D conversion, it appears as a contrast

changing in the vertical direction (column direction) in an image output from the photoelectric conversion apparatus **100**. This is referred to as vertical shading. The vertical shading is corrected by the signal processing unit **108**.

(48) The first embodiment provides a function of removing or reducing the influence of vertical shading in the photoelectric conversion apparatus **100** that includes the readout circuit **105** which reads out signals from the pixel array **101** with a gain selected from a plurality of gains. With reference to FIGS. **7** to **9**, the arrangement and operation of the photoelectric conversion apparatus **100** according to the first embodiment will be described below.

(49) FIG. **7** exemplarily shows the arrangement of the pixel array **101**. The pixel array **101** includes an effective pixel region **702** for generating a pixel signal or an image signal by photoelectric conversion. The effective pixel region **702** is a region where the plurality of pixels (effective pixels) **102** are arranged so as to form the plurality of rows and the plurality of columns. The pixel array **101** can further include a first reference pixel region **703** and a second reference pixel region **704**. In the reference pixel regions **703** and **704**, a plurality of reference pixels, each of which outputs a noise level, can be arranged. In the first reference pixel region **703**, a plurality of reference pixels can be arranged so as to form at least one column, and preferably a plurality of columns. Similarly, in the second reference pixel region **704**, a plurality of reference pixels can be arranged so as to form at least one column, and preferably a plurality of columns. The reference pixel can have an arrangement obtained by, for example, removing the photoelectric conversion element from the pixel **102**. The reference pixel is shielded from light by a light shielding film. The reference pixel can be used to generate a correction value for removing or reducing the influence of vertical shading described above. The first reference pixel region **703** can be used to generate the first correction value for correcting the pixel signal output from the pixel **102** which has output an optical signal level smaller than the determination level V_s , and having undergone a CDS process. The second reference pixel region **704** can be used to generate the second correction value for correcting the pixel signal output from the pixel **102** which has output an optical signal level larger than the determination level V_s , and having undergone a CDS process. The pixel array **101** can further include an Optical Black (OB) pixel region **701**. A plurality of OB pixels, each of which outputs an OB level, are arranged in the OB pixel region **701**. The OB pixel can be, for example, a pixel having the same arrangement as that of the pixel **102** but covered with a light shielding film. The OB pixel region **701** can include multiple OB pixels connected to the plurality of column signal lines **110** arranged in the effective pixel region **702**, and multiple OB pixels connected to the plurality of row selection lines **109** arranged in the effective pixel region **702**.

(50) In the photoelectric conversion apparatus **100** according to the first embodiment, a calibration operation for generating a correction value used to correct vertical shading, and an image capturing operation for generating an image by photoelectric conversion are executed. In the calibration operation, the readout circuit **105** reads out signals from the reference pixels in the reference pixel regions **703** and **704**, and the signal processing unit **108** processes the signals. Thus, the first and second correction values can be generated.

(51) In the first embodiment, the first correction value is generated by processing a signal obtained by reading out a noise level from the first reference pixel region **703** by the readout circuit **105**. Further, in the first embodiment, the second correction value is generated by processing a signal obtained by reading out a noise level from the second reference pixel region **704** by the readout circuit **105**.

(52) FIG. **8A** exemplarily shows an operation of reading out a signal from the reference pixel in the first reference pixel region **703** in the calibration operation. First, an operation of A/D-converting a predetermined voltage (here, a predetermined voltage V_{L1} for the sake of descriptive convenience) by the column circuit **400** of the readout circuit **105** is performed. More specifically, the operation of the comparator **401** is started at time t_{801} . At time t_{802} , the count of the counter/latch circuit **402** is reset, and the first reference signal $VRAMP-1$, which changes at the first change rate, is

supplied from the ramp circuit **403** to the comparator **401**. By performing a count operation in the period (time **t802** to time **t803**) from the reset of the count of the counter/latch circuit **402** to the inversion of the output of the comparator **401**, the predetermined voltage VL1 is A/D-converted. (53) Then, a predetermined voltage (here, a predetermined voltage VH1 for the sake of descriptive convenience) is A/D-converted. More specifically, at time **t804**, the count of the counter/latch circuit **402** is reset, and the first reference signal VRAMP-1 is supplied from the ramp circuit **403** to the comparator **401**. By performing the count operation in the period (time **t804** to time **t805**) from the reset of the count of the counter/latch circuit **402** to the inversion of the output of the comparator **401**, the predetermined voltage VH1 is A/D-converted.

(54) FIG. **8B** exemplarily shows an operation of reading out a signal from the reference pixel in the second reference pixel region **704** in the calibration operation. First, an operation of A/D-converting a predetermined voltage (here, a predetermined voltage VL2 for the sake of descriptive convenience) by the column circuit **400** of the readout circuit **105** is performed. More specifically, the operation of the comparator **401** is started at time **t806**. At time **t807**, the count of the counter/latch circuit **402** is reset, and the first reference signal VRAMP-1, which changes at the first change rate, is supplied from the ramp circuit **403** to the comparator **401**. By performing the count operation in the period (time **t807** to time **t808**) from the reset of the count of the counter/latch circuit **402** to the inversion of the output of the comparator **401**, the predetermined voltage VL2 is A/D-converted.

(55) Then, a predetermined voltage (here, a predetermined voltage VH2 for the sake of descriptive convenience) is A/D-converted. More specifically, at time **t809**, the count of the counter/latch circuit **402** is reset, and the second reference signal VRAMP-2 is supplied from the ramp circuit **403** to the comparator **401**. By performing the count operation in the period (time **t809** to time **t810**) from the reset of the count of the counter/latch circuit **402** to the inversion of the output of the comparator **401**, the predetermined voltage VH2 is A/D-converted. As has been described above, the second reference signal VRAMP-2 is a ramp signal which changes at the second change rate that is a times the first change rate of the first reference signal VRAMP-1.

(56) FIG. **9** exemplarily shows an arrangement example of the signal processing unit **108**. The signal processing unit **108** can include a CDS circuit **900** that performs a CDS process on a pair of digital signals provided from the pixel array **101** (effective pixel, reference pixel, or OB pixel) via the column signal line **110**, the readout circuit **105**, and the memory unit **106**. Here, in the image capturing operation, when reading out signals from the effective pixel, the pair of digital signals supplied to the CDS circuit **900** is composed of the digital signal of the noise level and the digital signal of the optical signal level. In the image capturing operation, when reading out signals from the OB pixel, the pair of digital signals supplied to the CDS circuit **900** is composed of the digital signal of the reset level and the digital signal of the dark level which is read out while setting the transfer transistor **202** in the ON state.

(57) In the calibration operation, signals are read out from the reference pixels in the first reference pixel region **703** and the second reference pixel region **704**. The readout circuit **105** can perform readout of the signal from the reference pixel in the first reference pixel region **703** and readout of the signal from the reference pixel in the second reference pixel region **704** in parallel, that is, simultaneously.

(58) First, a process of generating the first correction values for one column (n pieces) by reading out signals from the reference pixels in the first reference pixel region **703** will be described. As has been described with reference to FIG. **8A**, the pair of digital signals supplied to the CDS circuit **900** is composed of a first digital signal D1 read out with the first gain (first reference signal VRAMP-1) and a second digital signal D2 read out with the first gain. The first digital signal D1 can be expressed as $NL+DL(T1)$. The second digital signal D2 can be expressed as $NL+DL(T2)$.

Accordingly, an output of the CDS circuit **900** can be expressed as:

$$(NL+DL(T2))-(NL+DL(T1))=DL(T2)-DL(T1) \quad (3)$$

(59) The right-hand side of equation (3) corresponds to $DL(T2)-DL(T1)$ which is vertical shading to be removed in the right-hand side of equation (1). That is, when signals read out from the first reference pixel region **703** are input in the calibration operation, the CDS circuit **900** outputs $DL(T2)-DL(T1)$. $DL(T2)-DL(T1)$ can be used as the first correction value for correcting the pixel signal read out from the pixel **102** which has output an optical signal level smaller than the determination level V_s , and having undergone a CDS process by the CDS circuit **900**. In other words, $DL(T2)-DL(T1)$ can be used as the first correction value for correcting the pixel value read out from the pixel **102** which has output an optical signal level with the luminance determination information LL of 0, and having undergone the CDS process by the CDS circuit **900**.

(60) The signal processing unit **108** may include a holding unit **902**, and the holding unit **902** can include a first holding region **9021** for holding at least the first correction values for one column (n pieces). Here, multiple pairs of digital signals may be acquired by reading out signals multiple times from each reference pixel in the first reference pixel region **703**, and the average value of multiple first provisional correction values obtained by supplying the multiple pairs of digital signals to the CDS circuit **900** may be used as the first correction value. Alternatively, if the first reference pixel region **703** includes the plurality of columns of reference pixels, the average value of multiple first provisional correction values obtained from the plurality of columns of reference pixels may be used as the first correction value. Alternatively, a value obtained by performing filter processing on a plurality of first provisional correction values obtained from the plurality of reference pixels in the first reference pixel region **703** may be used as the first correction value. The CDS circuit **900** can also be understood as a generation circuit that generates the first correction value.

(61) Next, a process of generating the second correction values for one column (n pieces) by reading out signals from the reference pixels in the second reference pixel region **704** will be described. As has been described with reference to FIG. **8B**, the pair of digital signals is composed of the first digital signal **D1** read out with the first gain (first reference signal VRAMP-1) and the second digital signal **D2** read out with the second gain (second reference signal VRAMP-2). The first digital signal **D1** can be expressed as $NL+DL(T1)$. The second digital signal **D2** can be expressed as $NL+DH(T3)$. Accordingly, an output of the CDS circuit **900** can be expressed as:

$$\alpha \times (NL+DH(T3)) - (NL+DL(T1)) = \alpha DH(T3) - DL(T1) \quad (4)$$

(62) The right-hand side of equation (4) corresponds to $\alpha DH(T3)-DL(T1)$ which is vertical shading to be removed in the right-hand side of equation (2). That is, when signals read out from the second reference pixel region **704** are input in the calibration operation, the CDS circuit **900** outputs $\alpha DH(T3)-DL(T1)$. $\alpha DH(T3)-DL(T1)$ can be used as the second correction value for correcting the pixel signal read out from the pixel **102** which has output an optical signal level larger than the determination level V_s , and having undergone a CDS process by the CDS circuit **900**. In other words, $\alpha DH(T3)-DL(T1)$ can be used as the second correction value for correcting the pixel value read out from the pixel **102** which has output an optical signal level with the luminance determination information LL of 1, and having undergone the CDS process by the CDS circuit **900**.

(63) The holding unit **902** of the signal processing unit **108** can include a second holding region **9022** for holding at least the second correction values for one column (n pieces). Here, multiple pairs of digital signals may be acquired by reading out signals multiple times from each reference pixel in the second reference pixel region **704**, and the average value of multiple second provisional correction values obtained by supplying the multiple pairs of digital signals to the CDS circuit **900** may be used as the second correction value. Alternatively, if the second reference pixel region **704** includes the plurality of columns of reference pixels, the average value of multiple second provisional correction values obtained from the plurality of columns of reference pixels may be used as the second correction value. Alternatively, a value obtained by performing filter processing on a plurality of second provisional correction values obtained from the plurality of reference pixels in the second reference pixel region **704** may be used as the second correction value. The

CDS circuit **900** can also be understood as a generation circuit that generates the second correction value.

(64) The signal processing unit **108** can include a correction unit **1000** that corrects, using the first correction value or the second correction value provided from the holding unit **902**, the signal (corresponding to the right-hand side of equation (1) or equation (2)) output from the CDS circuit **900** in the image capturing operation. The correction unit **1000** can include a selection unit **1001** and a subtractor **1002**. In the image capturing operation, the CDS circuit **900** is provided with the pair of digital signals read out from the effective pixel or OB pixel via the column signal line **110** by the readout circuit **105** and stored in the memory unit **106**, and the luminance determination information LL. The CDS circuit **900** supplies, to the correction unit **1000**, a signal (corresponding to the right-hand side of equation (1) or equation (2)) Dcds obtained by performing the CDS process on the pair of digital signals, and the signal Dcds is supplied to the subtractor **1002**.

(65) The holding unit **902** can supply, to the selection unit **1001**, the first correction value (corresponding to the right-hand side of equation (3)) corresponding to the pair of digital signals among a plurality of first correction values stored in the first holding region **9021**. The holding unit **902** can also supply, to the selection unit **1001**, the second correction value (corresponding to the right-hand side of equation (4)) corresponding to the pair of digital signals among a plurality of second correction values stored in the second holding region **9022**. If the luminance determination information LL corresponding to the pair of digital signals is 0, the selection unit **1001** supplies, to the subtractor **1002**, the first correction value supplied from the first holding region **9021**. If the luminance determination information LL is 1, the selection unit **1001** supplies, to the subtractor **1002**, the second correction value supplied from the second holding region **9022**. The correction unit **1000** performs calculation of subtracting, from the signal Dcds supplied from the CDS circuit **900**, the first or second correction value supplied from the selection unit **1001**, and outputs the result as a pixel signal DO. This calculation can be expressed as equation (55) if the luminance determination information LL is 0, and can be expressed as equation (B6) if the luminance determination information LL is 1.

$$SL+DL(T2)-DL(T1)-(DL(T2)-DL(T1))=SL \quad (5)$$

$$\alpha SH+\alpha DH(T3)-DL(T1)-(\alpha DH(T3)-DL(T1))=\alpha SH \quad (6)$$

(66) That is, for the CDS process (correlated double sampling), the readout circuit **105** reads out the noise level with the first gain from the effective pixel selected from the plurality of effective pixels, and reads out the optical signal level with the first gain or the second gain. In shading correction, the pixel signal of the effective pixel from which the optical signal level has been read out with the first gain is corrected based on the first correction value, and the pixel signal of the effective pixel from which the optical signal level has been read out with the second gain is corrected based on the second correction value.

(67) Vertical shading can change depending on temperature. Therefore, it is preferable to execute the calibration operation at an arbitrary timing or a timing planned in advance. Alternatively, the first correction value and the second correction value may be generated in each frame (vertical scanning period), and the signal Dcds may be corrected based on the first correction value and the second correction value in the same frame (vertical scanning period). The first correction value and the second correction value may be generated outside the photoelectric conversion apparatus **100**, and provided to the photoelectric conversion apparatus **100** or the holding unit **902**.

(68) With reference to FIGS. **10** and **11**, the arrangement and operation of a photoelectric conversion apparatus **100** according to the second embodiment will be described below. Matters not mentioned as the second embodiment can follow the first embodiment. The photoelectric conversion apparatus **100** according to the second embodiment has an arrangement obtained by removing the first reference pixel region **703** and the first holding region **9021** from the photoelectric conversion apparatus **100** according to the first embodiment, and replacing the first correction value with a fixed value. The second embodiment is useful in a case in which

DL(T2)–DL(T1) in equation (1) described in the first embodiment is negligible. In other words, the photoelectric conversion apparatus **100** of the second embodiment does not correct vertical shading if luminance determination information LL is 0.

(69) As exemplarily shown in FIG. **10**, in the second embodiment, the first reference pixel region **703** and the second reference pixel region **704** in the first embodiment are replaced with a reference pixel region **705**. The reference pixel region **705** in the second embodiment of the third disclosure corresponds to the second reference pixel region **704** in the first embodiment. FIG. **11** shows an arrangement example of a signal processing unit **108** of the second embodiment. The signal processing unit **108** can include a CDS circuit **900** that performs a CDS process on a pair of digital signals provided from a pixel array **101** (effective pixel, reference pixel, or OB pixel) via a column signal line **110**, a readout circuit **105**, and a memory unit **106**. Here, in an image capturing operation, when reading out signals from the effective pixel, the pair of digital signals supplied to the CDS circuit **900** is composed of the digital signal of the noise level and the digital signal of the optical signal level. In the image capturing operation, when reading out signals from the OB pixel, the pair of digital signals is composed of the digital signal of the reset level and the digital signal of the dark level which is read out while setting a transfer transistor **202** in the ON state.

(70) In a calibration operation, correction values for one column (n pieces) are generated by reading out signals from the reference pixels in the reference pixel region **705**. As has been described with reference to FIG. **8B**, the pair of digital signals supplied to the CDS circuit **900** in the calibration operation is composed of a first digital signal D1 read out with the first gain and a second digital signal D2 read out with the second gain. The first gain corresponds to a first reference signal VRAMP-1, and the second gain corresponds to a second reference signal VRAMP-2. The first digital signal D1 can be expressed as $NL+DL(T1)$. The second digital signal D2 can be expressed as $NL+DH(T3)$. Accordingly, an output of the CDS circuit **900** can be expressed as equation (4) described above. An output of the CDS circuit **900** in the calibration operation can be used as the correction value for correcting the pixel signal read out from a pixel **102** which has output an optical signal level with the luminance determination information LL of 1, and having undergone a CDS process by the CDS circuit **900**. A holding unit **902** of the signal processing unit **108** can include a holding region **9022** for holding at least the correction values for one column (n pieces).

(71) In the image capturing operation, a correction unit **1000** of the signal processing unit **108** corrects the signal (corresponding to the right-hand side of equation (1) or equation (2)) output from the CDS circuit **900** using the fixed value or the second correction value from the holding unit **902**. The correction unit **1000** can include a selection unit **1001** and a subtractor **1002**. In the image capturing operation, the CDS circuit **900** is provided with a pair of digital signals read out from the effective pixel or OB pixel via the column signal line **110** by the readout circuit **105** and stored in the memory unit **106**, and the luminance determination information LL. The CDS circuit **900** supplies, to the correction unit **1000**, a signal (corresponding to the right-hand side of equation (1) or equation (2)) Dcds obtained by performing the CDS process on the pair of digital signals, and the signal Dcds is supplied to the subtractor **1002**. The holding unit **902** can supply, to the selection unit **1001**, the correction value (corresponding to the right-hand side of equation (4)) corresponding to the pair of digital signals among a plurality of correction values (correction values for one column) stored in the holding region **9022**. If the luminance determination information LL corresponding to the pair of digital signals is 0, the selection unit **1001** can supply the fixed value to the subtractor **1002**. If the luminance determination information LL is 1, the selection unit **1001** can supply, to the subtractor **1002**, the correction value supplied from the holding region **9022**. The correction unit **1000** performs calculation of subtracting, from the signal Dcds supplied from the CDS circuit **900**, the fixed value or correction value supplied from the selection unit **1001**, and outputs the result as a pixel signal DO. This calculation can be expressed as equation (7) if the luminance determination information LL is 0, and can be expressed as equation (8) if the

luminance determination information LL is 1.

$$SL+DL(T2)-DL(T1)-(fixed\ value) \quad (7)$$

$$\alpha SH+\alpha DH(T3)-DL(T1)-(\alpha DH(T3)-DL(T1))=\alpha SH \quad (8)$$

(72) A photoelectric conversion apparatus **100** according to the third embodiment of the third disclosure will be described below. Matters not mentioned as the third embodiment of the third disclosure can follow the first or second embodiment of the third disclosure. In the photoelectric conversion apparatus **100** according to the third embodiment, the pixel array **101** has an arrangement as exemplarily shown in FIG. **10** described in the second embodiment, that is, an arrangement including a single reference pixel region **705**. In the third embodiment, a first correction value and a second correction value are generated as in the first embodiment using the single reference pixel region **705**.

(73) In a calibration operation in the third embodiment, a first operation of reading out signals from the reference pixel region **705** to generate the first correction value and a second operation of reading out signals from the reference pixel region **705** to generate the second correction value are performed in periods different from each other.

(74) First, a first process of generating the first correction values for one column (n pieces) by reading out signals from the reference pixels in the reference pixel region **705** by a readout circuit **105** will be described. In the first process, as exemplarily shown in FIG. **8A**, signals are read out from the reference pixel in the reference pixel region **705** by the readout circuit **105**, and a pair of digital signals is supplied to a CDS circuit **9900**. The pair of digital signals is composed a first digital signal **D1** read out from the reference pixel in the reference pixel region **705** with a first gain (first reference signal VRAMP-1), and a second digital signal **D2** read out from the reference pixel in the reference pixel region **705** with the first gain. The first digital signal **D1** can be expressed as $NL+DL(T1)$. The second digital signal **D2** can be expressed as $NL+DL(T2)$. Accordingly, an output of the CDS circuit **900** corresponds to $DL(T2)-DL(T1)$ as expressed by equation (3) described above. $DL(T2)-DL(T1)$ can be used as the first correction value for correcting the pixel signal read out from a pixel **102** which has output an optical signal level smaller than a determination level V_s , and having undergone a CDS process by the CDS circuit **900**. In other words, $DL(T2)-DL(T1)$ can be used as the first correction value for correcting the pixel value read out from the pixel **102** which has output an optical signal level with luminance determination information LL of 0, and having undergone the CDS process by the CDS circuit **900**. A holding unit **902** of a signal processing unit **108** can include a first holding region **9021** for holding at least the first correction values for one column (n pieces). Here, multiple pairs of digital signals may be acquired by reading out signals multiple times from each reference pixel in the reference pixel region **705**, and the average value of multiple first provisional correction values obtained by supplying the multiple pairs of digital signals to the CDS circuit **900** may be used as the first correction value. Alternatively, if the reference pixel region **705** includes a plurality of columns of reference pixels, the average value of multiple first provisional correction values obtained from the plurality of columns of reference pixels may be used as the first correction value. Alternatively, a value obtained by performing filter processing on a plurality of first provisional correction values obtained from the plurality of reference pixels in the reference pixel region **705** may be used as the first correction value. The CDS circuit **900** can also be understood as a generation circuit that generates the first correction value.

(75) Next, a second process of generating the second correction values for one column (n pieces) by reading out signals from the reference pixels in the reference pixel region **705** by the readout circuit **105** will be described. In the second process, as exemplarily shown in FIG. **8B**, signals are read out from the reference pixel in the reference pixel region **705** by the readout circuit **105**, and a pair of digital signals is supplied to the CDS circuit **900**. The pair of digital signals is composed of the first digital signal **D1** read out with the first gain (first reference signal VRAMP-1), and the second digital signal **D2** readout with a second gain (second reference signal VRAMP-2). The first

digital signal D1 can be expressed as $NL+DL(T1)$. The second digital signal D2 can be expressed as $NL+DH(T3)$. Accordingly, an output of the CDS circuit **900** corresponds to $\alpha DH(T3)-DL(T1)$ as expressed by equation (4) described above. $\alpha DH(T3)-DL(T1)$ can be used as the second correction value for correcting the pixel signal read out from the pixel **102** which has output an optical signal level larger than the determination level V_s , and having undergone a CDS process by the CDS circuit **900**. In other words, $\alpha DH(T3)-DL(T1)$ can be used as the second correction value for correcting the pixel value read out from the pixel **102** which has output an optical signal level with the luminance determination information LL of 1, and having undergone the CDS process by the CDS circuit **900**. The holding unit **902** of the signal processing unit **108** can include a second holding region **9022** for holding at least the second correction values for one column (n pieces). Here, multiple pairs of digital signals may be acquired by reading out signals multiple times from each reference pixel in the reference pixel region **705**, and the average value of multiple second provisional correction values obtained by supplying the multiple pairs of digital signals to the CDS circuit **900** may be used as the second correction value. Alternatively, if the reference pixel region **705** includes a plurality of columns of reference pixels, the average value of multiple second provisional correction values obtained from the plurality of columns of reference pixels may be used as the second correction value. Alternatively, a value obtained by performing filter processing on a plurality of second provisional correction values obtained from the plurality of reference pixels in the reference pixel region **705** may be used as the second correction value. The CDS circuit **900** can also be understood as a generation circuit that generates the second correction value.

(76) The operation of a correction unit **1000** in an image capturing operation in the third embodiment is similar to that in the first embodiment.

(77) If the size of the reference pixel region is the same, the time required for generating the first and second correction values is longer in the third embodiment than in the first embodiment, but the number of data that can be used to generate the first and second correction values is larger in the third embodiment than in the first embodiment. Therefore, the third embodiment is advantageous in reducing the influence of random noise.

(78) With reference to FIG. **12**, the arrangement and operation of a photoelectric conversion apparatus **100** according to the fourth embodiment will be described. Matters not mentioned as the fourth embodiment can follow the first to third embodiments of the third disclosure. The fourth embodiment provides an example of the photoelectric conversion apparatus **100** that performs, in addition to correction of vertical shading described above, correction based on signals read out by a readout circuit **105** from a plurality of OB pixels arranged in an OB pixel region **701**. In one aspect, the vertical shading described in the first to third embodiments of the third disclosure occurs in A/D conversion. Further, correction based on signals read out from the plurality of OB pixels can be understood as correction of noise components (for example, vertical shading, horizontal shading, fixed pattern noise, and the like) generated in a pixel array **101**. Each reference pixel in reference pixel regions **703** and **704** outputs a fixed value or a predetermined voltage to a column signal line **110**. On the other hand, each OB pixel in the OB pixel region **701** outputs a signal or voltage corresponding to its characteristics to the column signal line **110**. That is, the signals or voltages output from the plurality of OB pixels in the OB pixel region **701** to the column signal lines **110** have variations reflecting the variations in characteristics among the plurality of OB pixels.

(79) FIG. **12** exemplarily shows the arrangement of a signal processing unit **108** of the fourth embodiment. The signal processing unit **108** can perform, in addition to shading correction, by a subtractor **1002**, on a signal D_{cds} output from a CDS circuit **900**, correction based on signals read out by the readout circuit **105** from the plurality of OB pixels arranged in the OB pixel region **701**. A correction unit **1000** can include, for example, an OB clamp processing unit **1003** in addition to a selection unit **1001** and the subtractor **1002** for correcting the signal D_{cds} based on a correction value generated using a plurality of reference pixels in the reference pixel regions **703** and **704**. The

OB clamp processing unit **1003** can be, for example, arranged so as to process the signal output from the subtractor **1002**, but may be arranged between the CDS circuit **900** and the subtractor **1002**. A holding unit **902** can include a third holding region **9023** for holding OB data to be supplied to the OB clamp processing unit **1003**. The OB data can be, for example, data read out from the plurality of OB pixels in the OB pixel region **701** by the readout circuit **105** and having undergone a CDS process by the CDS circuit **900**, or data obtained by further processing the CDS-processed data. The OB clamp processing unit **1003** performs an OB clamp process on the pixel signal output from the subtractor **1002** based on the data supplied from the third holding region **9023** in accordance with the position of a pixel **102** related to the signal Dcds (pixel signal), and outputs the result as a pixel signal DO.

(80) With reference to FIGS. **13** to **15B**, a photoelectric conversion apparatus **100** according to the fifth embodiment will be described below. Matters not mentioned as the fifth embodiment can follow the first to fourth embodiments of the third disclosure. FIG. **13** shows an arrangement example of one column circuit **400'** of a plurality of column circuits **400'** of a readout circuit **105** in the photoelectric conversion apparatus **100** according to the fifth embodiment. In the fifth embodiment, the column circuit **400** in the first embodiment is replaced with the column circuit **400'**. The column circuit **400'** includes an amplification circuit **404** that amplifies a signal VIN supplied from a pixel array **101** via a column signal line **110**. An output VA of the amplification circuit **404** is supplied to a comparator **401**. Such an arrangement is advantageous in expansion of the dynamic range and improvement of the S/N ratio.

(81) FIG. **14** exemplarily shows the operation of each column circuit **400'** in the readout circuit **105** in the fifth embodiment. In FIG. **14**, the abscissa exemplarily represents time, the ordinate in the upper stage exemplarily represents the level of the reference signal and the level of the signal from the pixel array, and the ordinate in the lower stage exemplarily represents the output of the comparator **401**. With reference to FIG. **14**, an example of changing the gain of the amplification circuit **404** in accordance with the level of the signal VIN from the pixel array **101** will be described. An operation of reading out a noise level and an operation of determining an optical signal level are similar to those in the period from time **t501** to time **t506** in FIG. **5**, so that a description thereof will be omitted. Note that in the period from time **t501** to time **t506**, the gain of the amplification circuit **404** is set to a first gain.

(82) If the optical signal level is smaller than a determination level V_s ($S < V_s$), A/D conversion is performed while keeping the gain of the amplification circuit **404** at the first gain. That is, as in the noise level readout period, A/D conversion of the optical signal level is performed from time **t507** while a ramp circuit **403** supplies a first reference signal VRAMP-1 to the comparator **401**. With this, in the example shown in FIG. **14**, a count value is obtained in the period from time **t507** to time **t508**. On the other hand, if the optical signal level is larger than the determination level V_s ($S > V_s$), the gain of the amplification circuit **404** is changed from the first gain to a second gain at time **t1101**, and then A/D conversion of the optical signal level is performed using the first reference signal VRAMP-1. With this, in the example shown in FIG. **14**, a count value can be obtained in the period from time **t507** to time **t1102**. With this, it is possible to implement expansion of the dynamic range and improvement of the S/N ratio.

(83) As has been described above, in the fifth embodiment, readout of the noise level and readout of the optical signal level are performed while performing A/D conversion using the reference signal having the same change rate. However, in the fifth embodiment, the amplification circuit **404** reads out the noise level with the first gain, and reads out the optical signal level with the second gain. Therefore, even if a CDS circuit **900** performs a CDS process using the noise level and the optical signal level, the difference between the gains of the amplification circuit **404** (for example, the difference between the capacitance values concerning the gains) can cause an offset error. The fifth embodiment provides a function of removing or reducing an offset error that can occur, in the configuration in which the amplification circuit **404** amplifies the signal from the pixel array **101**

with the gain selected from a plurality of gains, when amplifying the optical signal level and the noise level with different gains.

(84) Like the photoelectric conversion apparatus **100** according to the first embodiment, the photoelectric conversion apparatus **100** according to the fifth embodiment executes a calibration operation for generating a correction value used to correct vertical shading, and an image capturing operation for generating an image by photoelectric conversion. In the calibration operation, the readout circuit **105** reads out signals from the reference pixels in reference pixel regions **703** and **704**, and a signal processing unit **108** processes the signals. Thus, the correction value can be generated.

(85) FIG. **15A** exemplarily shows an operation of reading out a signal from the reference pixel in the first reference pixel region **703** in the calibration operation. First, an operation of A/D-converting a predetermined voltage (here, a predetermined voltage **VL1** for the sake of descriptive convenience) by the column circuit **400** of the readout circuit **105** is performed. More specifically, the operation of the comparator **401** is started at time **t1201**. At time **t1202**, the count of a counter/latch circuit **402** is reset, and the first reference signal **VRAMP-1**, which changes at the first change rate, is supplied from the ramp circuit **403** to the comparator **401**. The amplification circuit **404** outputs the voltage obtained by amplifying the predetermined voltage **VL1** with the first gain, and an A/D conversion unit (**401**, **402**, and **403**) A/D-converts the amplified voltage. By performing a count operation in the period (time **t1202** to time **t1203**) from the reset of the count of the counter/latch circuit **402** to the inversion of the output of the comparator **401**, the amplification operation and A/D conversion of the predetermined voltage **VL1** are performed.

(86) Then, the amplification operation and A/D conversion of a predetermined voltage (here, a predetermined voltage **VH1** for the sake of descriptive convenience) are performed. Here, the predetermined voltage **VH1** is amplified with the first gain by the amplification circuit **404** (time **t1204**), and then A/D conversion of the amplified voltage is performed. By performing the count operation in the period from time **t1205** to time **t1206**, the amplification operation and A/D conversion of the predetermined voltage **VH1** are performed.

(87) A first digital signal **D1** obtained by reading out the predetermined voltage **VL1** as described above and a second digital signal **D2** obtained by reading out the predetermined voltage **VH1** are supplied to the CDS circuit **900** of the signal processing unit **108** exemplarily shown in FIG. **9**. In the CDS circuit **900**, the first correction value used to correct the offset error is generated by the CDS process, and stored in a first holding region **9021** of a holding unit **902**.

(88) FIG. **15B** exemplarily shows an operation of reading out a signal from the reference pixel in the second reference pixel region **704** in the calibration operation. First, an operation of A/D-converting a predetermined voltage (here, a predetermined voltage **VL2** for the sake of descriptive convenience) by the column circuit **400** of the readout circuit **105** is performed. More specifically, the operation of the comparator **401** is started at time **t1201**. At time **t1202**, the count of the counter/latch circuit **402** is reset, and the first reference signal **VRAMP-1**, which changes at the first change rate, is supplied from the ramp circuit **403** to the comparator **401**. The amplification circuit **404** outputs the voltage obtained by amplifying the predetermined voltage **VL2** with the first gain, and the A/D conversion unit (**401**, **402**, and **403**) A/D-converts the amplified voltage. By performing the count operation in the period (time **t1202** to time **t1203**) from the reset of the count of the counter/latch circuit **402** to the inversion of the output of the comparator **401**, the amplification operation and A/D conversion of the predetermined voltage **VL2** are performed.

(89) Then, the amplification operation and A/D conversion of a predetermined voltage (here, a predetermined voltage **VH2** for the sake of descriptive convenience) are performed. Here, the predetermined voltage **VH2** is amplified with the second gain by the amplification circuit **404** (time **t1204**), and then A/D conversion of the amplified voltage is performed. By performing the count operation in the period from time **t1205** to time **t1207**, the amplification operation and A/D conversion of the predetermined voltage **VH2** are performed.

(90) The first digital signal D1 obtained by reading out the predetermined voltage VL2 as described above and the second digital signal D2 obtained by reading out the predetermined voltage VH2 are supplied to the CDS circuit 900 of the signal processing unit 108 exemplarily shown in FIG. 9. In the CDS circuit 900, the second correction value used to correct the offset error is generated by the CDS process, and stored in a second holding region 9022 of the holding unit 902.

(91) In the image capturing operation, a correction unit 1000 of the signal processing unit 108 corrects, using the first correction value or the second correction value provided from the holding unit 902, the signal output from the CDS circuit 900. In the image capturing operation, the CDS circuit 900 is provided with a pair of digital signals read out from the effective pixel or OB pixel via the column signal line 110 by the readout circuit 105 and stored in a memory unit 106, and luminance determination information LL. The CDS circuit 900 supplies, to the correction unit 1000, a signal Dcds obtained by performing the CDS process on the pair of digital signals, and the signal Dcds is supplied to a subtractor 1002. The holding unit 902 can supply, to a selection unit 1001, the first correction value corresponding to the pair of digital signals among a plurality of first correction values stored in the first holding region 9021. The holding unit 902 can also supply, to the selection unit 1001, the second correction value corresponding to the pair of digital signals among a plurality of second correction values stored in the second holding region 9022. If the luminance determination information LL corresponding to the pair of digital signals is 0, the selection unit 1001 supplies, to the subtractor 1002, the first correction value supplied from the first holding region 9021. If the luminance determination information LL is 1, the selection unit 1001 supplies, to the subtractor 1002, the second correction value supplied from the second holding region 9022. The correction unit 1000 performs calculation of subtracting, from the signal Dcds supplied from the CDS circuit 900, the first or second correction value supplied from the selection unit 1001, and outputs the result as a pixel signal DO.

(92) Note that in the fifth embodiment, the column circuit 400' includes the A/D converter. However, the column circuit 400' may output the signal output from the amplification circuit 404 without performing A/D conversion.

(93) With reference to FIG. 16, a photoelectric conversion apparatus 100 according to the sixth embodiment will be described below. The photoelectric conversion apparatus 100 according to the sixth embodiment can have an arrangement similar to that of the photoelectric conversion apparatus 100 according to the fifth embodiment. Matters not mentioned as the sixth embodiment can follow the fifth embodiment. In the sixth embodiment, in order to expand the dynamic range, optical signal levels are read out while time-sequentially switching the gain of an amplification circuit 404.

(94) FIG. 16 exemplarily shows the operation of one column circuit 400' in a readout circuit 105 in the sixth embodiment. Here, the operation of reading out signals from a pixel 102 in the jth row by a vertical scanning unit 103 is representatively shown. Note that each signal shown in FIG. 16 is a high-active signal. The operation of reading out the noise level and the optical signal level with the same gain is common to the operation in the period from time t301 to time t309 in FIG. 3. In FIG. 16, in the period from time t301 to time t309, the gain of the amplification circuit A404 is set at the first gain, and this is exemplarily shown as " $\times 1$ ".

(95) At time t1301, the gain of the amplification circuit 404 is changed from the first gain (for example, $\times 1$) to the second gain (for example, $\times 1/\alpha$) different from the first gain. Here, in order to change the gain of the amplification circuit 404, the feedback capacitance in the amplification circuit is changed. However, another configuration may also be used.

(96) In the period from time t1302 to time t1303, the optical signal level output to a column signal line 110 is multiplied by $1/\alpha$ by the amplification circuit 404, and then converted into a digital signal by the A/D conversion unit (401, 402, and 403). This operation is referred to as S conversion 2. In this manner, N conversion, S conversion, and S conversion 2 are performed, and each result is stored in a memory unit 106. Here, let N be the digital signal obtained by N conversion, S be the

digital signal obtained by S conversion, and S2 be the digital signal obtained by S conversion 2.

(97) In the period from time t**1304** to time t**1305**, signals of the pixels **102** (the signal of each pixel is composed of N, S, and S2) for one row stored in the memory unit **106** can be sequentially selected by a horizontal scanning unit **107**, and supplied to a signal processing unit **108**. By performing the horizontal scanning until the last column, readout of signals (N, S, S2) of the pixels **102** for one row arranged in the readout target row is completed. The signal processing unit **108** includes a CDS circuit **900** that performs a CDS process of generating a pixel signal by subtracting the noise level from the optical signal level. The CDS circuit **900** outputs a pixel signal Dcds having undergone the CDS process.

(98) The CDS circuit **900** of the signal processing unit **108** reduces the fixed pattern noise by performing a process of subtracting N from S and a process of subtracting N from S2. The dynamic range can be expanded using the signal obtained by subtracting N from S and the signal obtained by subtracting N from S2. However, since S2 and N are signals read out with gains different from each other, even if the process of subtracting N from S2 is performed, an offset error can remain. This offset error can appear as vertical stripes in an image output from the photoelectric conversion apparatus **100**.

(99) The correction value for correcting the offset error is acquired by a calibration operation similar to that in the fifth embodiment, and stored in a holding unit **902**. In an image capturing mode, the signal obtained by subtracting N from S2 by the CDS circuit **900** can be corrected based on the correction value.

(100) FIG. **17** shows an example of the structure of the photoelectric conversion apparatus **100** exemplarily described through the first to sixth embodiments of the third disclosure. The photoelectric conversion apparatus **100** may include a first substrate **1401** and a second substrate **1402**. Each of the first substrate **1401** and the second substrate **1402** can be formed by processing, for example, a semiconductor substrate such as a silicon substrate. On the first substrate **1401**, for example, the signal processing unit **108**, the vertical scanning unit **103**, a control unit **104**, the readout circuit **105**, the memory unit **106**, the horizontal scanning unit **107**, and the like can be arranged. On the second substrate **1402**, for example, a pixel array **101** including an OB pixel region **701**, an effective pixel region **702**, and a reference pixel region **703** can be arranged. As exemplarily shown in FIG. **17**, at least a part of the first substrate **1401** and at least a part of the second substrate **1402** can be stacked. With the arrangement as described above, when manufacturing the photoelectric conversion apparatus **100**, it is possible to apply a process suitable for each of an analog portion including the pixel array **101** and a logic portion including the signal processing unit **108**.

(101) FIG. **18** exemplarily shows the arrangement of an image capturing apparatus **1500** incorporating the photoelectric conversion apparatus **100**. The image capturing apparatus **1500** can include an image capturing unit **1501**, a signal correction unit **1502**, a CPU **1503**, an external input unit **1504**, an optical system **1505**, a video display unit **1506**, a recording unit **1507**, and a drive system **1508**. In an example, the image capturing unit **1501** is a functional block which includes the pixel array **101**, the vertical scanning unit **103**, the control unit **104**, the readout circuit **105**, the memory unit **106**, and the horizontal scanning unit **107**. Such the image capturing unit **1501** may be understood as a photoelectric conversion apparatus. Alternatively, the image capturing unit **1501** may be the photoelectric conversion apparatus **100** including the signal processing unit **108** described above. In an example, the signal correction unit **1502** can include the signal processing unit **108** described above.

(102) In the pixel array **101** of the image capturing unit **1501**, an optical image is formed by the optical system **1505**. The image capturing unit **1501** outputs an image signal corresponding to the optical image by photoelectric conversion. The signal correction unit **1502** corrects the image signal output from the image capturing unit **1501**, and outputs the corrected image signal to the video display unit **1506** and/or the recording unit **1507**. The CPU **1503** controls the respective

components in the image capturing unit **1500**. The drive system **1508** can, for example, drive the focusing lens of the optical system **1505** and drive the aperture. The external input unit **1504** can include various kinds of buttons and the like used by a user to input an image capturing condition and perform a shutter operation. The video display unit **1506** may be a touch panel. In this case, the video display unit **1506** can function as the external input unit **1504** (a part thereof).

(103) The present invention can also be implemented by executing the following process. That is, software (program) for implementing the above-described functions may be supplied to the system or apparatus via a network or various kinds of storage media. One or more processors (for example, a CPU and an MPU) in the computer of the system or apparatus can read out and execute the software (program). Alternatively, the above-described function can also be implemented by a circuit (for example, ASIC) having a fixed function, or a function definable circuit (for example, FPGA).

(104) While the present invention has been described with reference to exemplary embodiments, it is to be understood that the invention is not limited to the disclosed exemplary embodiments. The scope of the following claims is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures and functions.

(105) This application claims the benefit of Japanese Patent Application No. 2022-140208, filed Sep. 2, 2022, which is hereby incorporated by reference herein in its entirety.

Claims

1. A photoelectric conversion apparatus comprising a pixel array including a plurality of effective pixels, a readout circuit configured to read out a signal of the pixel array, and a signal processing unit configured to generate a pixel signal by performing correlated double sampling on signals read out by the readout circuit from an effective pixel selected from the plurality of effective pixels, and perform shading correction on the pixel signal, wherein the readout circuit has a function of reading out a signal of the pixel array with a first gain, and a function of reading out a signal of the pixel array with a second gain different from the first gain, the readout circuit reads out a noise level with the first gain and reads out an optical signal level with the first gain or the second gain from an effective pixel selected from the plurality of effective pixels for the correlated double sampling, and in the shading correction, a pixel signal of an effective pixel from which an optical signal level has been read out with the first gain is corrected based on a first correction value, and a pixel signal of an effective pixel from which an optical signal level has been read out with the second gain is corrected based on a second correction value.

2. The photoelectric conversion apparatus according to claim 1, further comprising a first holding unit configured to hold a plurality of first correction values respectively corresponding to a plurality of rows of the pixel array, and a second holding unit configured to hold a plurality of second correction values respectively corresponding to the plurality of rows, wherein the first correction value used in the shading correction is a first correction value obtained from the plurality of first correction values, and the second correction value used in the shading correction is a second correction value obtained from the plurality of second correction values.

3. The photoelectric conversion apparatus according to claim 1, wherein when reading out an optical signal level from a selected effective pixel, the readout circuit selects the first gain or the second gain in accordance with the optical signal level.

4. The photoelectric conversion apparatus according to claim 1, wherein the readout circuit includes an A/D converter, and the readout circuit performs A/D conversion using a first reference signal, which changes at a first change rate, when reading out a signal of the pixel array with the first gain, and performs A/D conversion using a second reference signal, which changes at a second change rate larger than the first change rate, when reading out a signal of the pixel array with the second gain.

5. The photoelectric conversion apparatus according to claim 2, wherein the pixel array includes a reference pixel region in which a plurality of reference pixels, each of which outputs a noise level, are arranged so as to form at least one column, each of the plurality of first correction values is generated by performing correlated double sampling, by the signal processing unit, on a signal obtained by reading out, by the readout circuit, a noise level with the first gain in a first period from a reference pixel in a selected row among the plurality of reference pixels, and a signal obtained by reading out a noise level with the first gain in a second period, and each of the plurality of second correction values is generated by performing correlated double sampling, by the signal processing unit, on a signal obtained by reading out, by the readout circuit, a noise level with the first gain in the first period from a reference pixel in the selected row among the plurality of reference pixels, and a signal obtained by reading out a noise level with the second gain in the second period.

6. The photoelectric conversion apparatus according to claim 2, wherein the pixel array includes a first reference pixel region in which a plurality of reference pixels, each of which outputs a noise level, are arranged so as to form at least one column, and a second reference pixel region in which a plurality of reference pixels, each of which outputs a noise level, are arranged so as to form at least one column, each of the plurality of first correction values is generated by performing correlated double sampling, by the signal processing unit, on a signal obtained by reading out, by the readout circuit, a noise level with the first gain in a first period from a reference pixel in a selected row among the plurality of reference pixels in the first reference pixel region, and a signal obtained by reading out a noise level with the first gain in a second period, and each of the plurality of second correction values is generated by performing correlated double sampling, by the signal processing unit, on a signal obtained by reading out, by the readout circuit, a noise level with the first gain in the first period from a reference pixel in the selected row among the plurality of reference pixels in the second reference pixel region, and a signal obtained by reading out a noise level with the second gain in the second period.

7. The photoelectric conversion apparatus according to claim 2, wherein the pixel array includes a reference pixel region in which a plurality of reference pixels, each of which outputs a noise level, are arranged so as to form at least one column, each of the plurality of first correction values is generated by performing correlated double sampling, by the signal processing unit, on a signal obtained by reading out, by the readout circuit, a noise level with the first gain in a first period from a reference pixel in a selected row among the plurality of reference pixels, and a signal obtained by reading out a noise level with the first gain in a second period, and each of the plurality of second correction values is generated by performing correlated double sampling, by the signal processing unit, on a signal obtained by reading out, by the readout circuit, a noise level with the first gain in a third period from the reference pixel in the selected row among the plurality of reference pixels, and a signal obtained by reading out a noise level with the second gain in a fourth period, and the first period, the second period, the third period, and the fourth period are periods different from each other.

8. The photoelectric conversion apparatus according to claim 1, wherein the readout circuit includes an A/D converter, and the readout circuit performs A/D conversion using a first reference signal, which changes at a first change rate, when reading out a signal of the pixel array with the first gain, and performs A/D conversion using a second reference signal, which changes at a second change rate larger than the first change rate, when reading out a signal of the pixel array with the second gain.

9. The photoelectric conversion apparatus according to claim 1, wherein the readout circuit includes an amplification circuit configured to amplify a signal output from the pixel array, and an A/D converter configured to A/D-convert a signal output from the amplification circuit, and a gain of the amplification circuit used when reading out a signal of the pixel array with the first gain and a gain of the amplification circuit used when reading out a signal of the pixel array with the second gain are different from each other.

10. The photoelectric conversion apparatus according to claim 1, wherein the pixel array includes a plurality of OB pixels each of which includes a photoelectric conversion element shielded from light, and the signal processing unit performs correction based on signals read out from the plurality of OB pixels by the readout circuit, in addition to performing the shading correction on the pixel signal.
11. A photoelectric conversion apparatus comprising a pixel array including a plurality of effective pixels and a reference pixel, a readout circuit configured to read out a signal of the pixel array as a digital signal, and a signal processing unit configured to generate a pixel signal by performing correlated double sampling on signals read out by the readout circuit from an effective pixel selected from the plurality of effective pixels, and perform shading correction on the pixel signal, wherein the readout circuit has a function of reading out a signal of the pixel array with a first gain, and a function of reading out a signal of the pixel array with a second gain different from the first gain, the readout circuit reads out a noise level with the first gain and reads out an optical signal level with the first gain or the second gain from an effective pixel selected from the plurality of effective pixels for the correlated double sampling, in the shading correction, a pixel signal of an effective pixel from which an optical signal level has been read out with the second gain is corrected based on a correction value, and the correction value is generated by performing correlated double sampling, by the signal processing unit, on a signal obtained by reading out a noise level with the first gain in a first period from the reference pixel by the readout circuit, and a signal obtained by reading out a noise level with the second gain in a second period.
12. A photoelectric conversion system that comprises a photoelectric conversion apparatus and a signal processing unit, wherein the photoelectric conversion apparatus comprises a pixel array including a plurality of effective pixels, and a readout circuit configured to read out a signal of the pixel array as a digital signal, the signal processing unit generates a pixel signal by performing correlated double sampling on signals read out by the readout circuit from an effective pixel selected from the plurality of effective pixels, and performs shading correction on the pixel signal, the readout circuit has a function of reading out a signal of the pixel array with a first gain, and a function of reading out a signal of the pixel array with a second gain different from the first gain, the readout circuit reads out a noise level with the first gain and reads out an optical signal level with the first gain or the second gain from an effective pixel selected from the plurality of effective pixels for the correlated double sampling, and in the shading correction, a pixel signal of an effective pixel from which an optical signal level has been read out with the first gain is corrected based on a first correction value, and a pixel signal of an effective pixel from which an optical signal level has been read out with the second gain is corrected based on a second correction value.
13. A photoelectric conversion system that comprises a photoelectric conversion apparatus and a signal processing unit, wherein the photoelectric conversion apparatus comprises a pixel array including a plurality of effective pixels and a reference pixel, and a readout circuit configured to read out a signal of the pixel array as a digital signal, the signal processing unit generates a pixel signal by performing correlated double sampling on signals read out by the readout circuit from an effective pixel selected from the plurality of effective pixels, and performs shading correction on the pixel signal, the readout circuit has a function of reading out a signal of the pixel array with a first gain, and a function of reading out a signal of the pixel array with a second gain different from the first gain, the readout circuit reads out a noise level with the first gain and reads out an optical signal level with the first gain or the second gain from an effective pixel selected from the plurality of effective pixels for the correlated double sampling, in the shading correction, a pixel signal of an effective pixel from which an optical signal level has been read out with the second gain is corrected based on a correction value, and the correction value is generated by performing correlated double sampling, by the signal processing unit, on a signal obtained by reading out a noise level with the first gain in a first period from the reference pixel by the readout circuit, and a signal obtained by reading out a noise level with the second gain in the second period.

14. A photoelectric conversion method comprising reading out, as a digital signal, a noise level of an effective pixel in a pixel array with a first gain, reading out, as a digital signal, an optical signal level of the effective pixel with the first gain or a second gain different from the first gain, generating a pixel signal by performing correlated double sampling on the digital signal of the noise level and the digital signal of the optical signal level, and a step of performing shading correction on the pixel signal, wherein in the shading correction, a pixel signal of an effective pixel from which an optical signal level has been read out with the first gain is corrected based on a first correction value, and a pixel signal of an effective pixel from which an optical signal level has been read out with the second gain is corrected based on a second correction value.

15. A photoelectric conversion method comprising reading out, as a digital signal, a noise level of an effective pixel in a pixel array with a first gain, reading out, as a digital signal, an optical signal level of the effective pixel with the first gain or a second gain different from the first gain, generating a pixel signal by performing correlated double sampling on the digital signal of the noise level and the digital signal of the optical signal level, and performing shading correction on the pixel signal, wherein in the shading correction, a pixel signal of an effective pixel from which an optical signal level has been read out with the second gain is corrected based on a correction value, and the correction value is generated by performing correlated double sampling on a signal obtained by reading out a noise level with the first gain in a first period from a reference pixel in the pixel array, and a signal obtained by reading out a noise level with the second gain in a second period.
